

Web accessibility guidelines. As per the two-year-old World Population Ageing UN report, “The number of older persons was 841 million in 2013, which is four times higher than the 202 million that lived in 1950. The older population will almost triple by 2050, when it is expected to surpass the two billion mark.” The growing population of the elderly further highlights the urgent need for accessible websites.

Let's try to understand the standards and guidelines that have been framed to improve accessibility. We can also look at the basic checks and some simple free tools which can be used to verify usability and accessibility.

Standards of accessibility

The international standards organisation, the WAI (Web Accessibility Initiative), which comes under the World Wide Web Consortium (W3C), defines standards for the WWW. The W3C has defined accessibility guidelines, based on which countries have derived their own standards. WCAG (Web Content Accessibility Guidelines) standards, framed by WAI are followed by each country as guidelines and country-specific laws are made as per the demographic requirements. The websites targeted at a specific country should be compliant with its defined standards.

The Web accessibility guidelines are defined for government websites, yet, till the date of writing this article, most public websites do not have the mandate to follow Web accessibility guidelines. Still, it is advisable to conform with country-specific laws. However, only in the USA has there been considerable litigation to make websites or products more accessible. Firms like Hilton Worldwide, Ticketmaster, Netflix and Ebay have settled disputes and agreed to make their websites more accessible.

The guidelines

- WCAG 2.0 — W3C Web Content Accessibility Guidelines 2.0
- WCAG 1.0 — W3C Web Content Accessibility Guidelines 1.0
- BITV - The German government's standards
- RGAA - The French government's standards
- JIS - The Japanese industry standard
- Irish National IT Accessibility Guidelines
- Section 508 – The US federal procurement standards
- Stanca Act - Italian accessibility legislation

The European Union follows the same guidelines as defined by WCAG2.0. Unfortunately, India has not defined specific laws for Web accessibility but all government sites follow the standard WCAG 2.0 guidelines.

Usability, accessibility and assistive technologies

Usability is ease of use and accessibility means access to all, irrespective of technological or physical barriers. A site or product can be usable but not accessible, and vice-versa.

The term ‘usability’ covers user satisfaction, the user experience and usefulness. Web designers should opt for a user-centred approach. As Jakob Nielsen, principal of the Nielsen Norman Group, and renowned Web usability consultant, rightly pointed out, “Studies of user behaviour on the Web find a low tolerance for difficult designs or slow sites. People don't want to wait. And they don't want to learn how to use a home page. There's no such thing as a training class or a manual for a website. People have to be able to grasp the functioning of the site immediately after scanning the home page — for a few seconds at most.” Usability, in terms of Web design, demands that the content be efficient, easy to understand and let the user reach the required data as fast as possible.

Accessibility literally implies the ability to access or as the dictionary states, ‘The quality of being at hand when needed.’ Accessible designs give users with disabilities, direct or indirect access to the content. Direct access implies that the user can access the content and can get the required information without any assistive technology devices. Accessibility advocates a universal design of the Web, which can be used by larger audiences across the globe while covering every possible range of disabilities. It may seem almost impossible to meet the diverse requirements for accessibility and usability design but, in turn, if the guidelines are studied and implemented properly in the design phase, the benefit is that the site addresses a larger user base.

Accessible design guidelines define a standard format that assistive technology developers use in assistive, adaptive, and rehabilitative devices to facilitate differently able users to comprehend and overcome their disability in some way or the other. The screen reader is an example of an assistive technology which converts the text on websites into speech. It is software that is installed on the PC and when we visit websites, the text is spoken out audibly for visually challenged users.

The screen reader comes as commercial or free versions. Some of the free versions of the screen reader can be downloaded from the links given below:

- <http://www.nvaccess.org/>
- <http://safa.sourceforge.net/>
- <http://www.satogo.com/en/>
- <http://www.screenreader.net/index.php?pageid=11>
- <http://webanywhere.cs.washington.edu/wa.php>

How a screen readers works

Screen readers interact with the Web browser using common Accessibility APIs which are built into the browser's applications. The irony is that, currently, there is no cross-platform accessibility API. So if the screen reader works perfectly on one browser, accessing the same website via another browser could mean an entirely different user experience, with some features missing.